

CS 613: Natural Language Processing

ASSIGNMENT 2

I. Report (50 Pts)

Write the feedback provided and the novelties discussed with the mentor (10 Pts)

- a. Write all feedback since Presentation 1, and report the work completed in weekly meetings with your mentors.

1. Report the other baselines you want to compete with (10 Pts)

- a. What is the state-of-the-art (the best paper that has solved the approach)?
- b. Are baseline implementations available for the novelties?

2. Datasets (05 Pts)

- a. How did you tackle the bias in your samples?
- b. What bottleneck did you encounter?
- c. How did you extend the existing dataset, and what are the differences between it and the previous one?

3. Experimental Plan (15 Pts)

- a. What experimental setting did you choose? How did you train the models? What methods did you use to ensure the results were robust, considering the limited computational resources?
- b. What models/approaches did you use?
 - i. How did you ensure that the approaches differed from one another?
 - ii. Compute the number of parameters and report the architectural differences.
- c. How to find the best params for your algos and baselines?
 - i. Grid search or better techniques.
 - ii. Prepare the ablation table with different hyperparameters.
- d. What metric did you choose? And why?
 - i. What additional metrics did you try with, other than the existing metrics?
- e. What level of system effort was required? Please paint a scenario for the final demo that you will present during the viva.

4. Project Management (10 Pts)

- a. How did you manage to propose the novel solutions?
 - b. Computational resources (RAM, GPUs, CPU Cores, Hard Disk, etc.) are needed.
 - c. How will you divide the project among yourselves? What tasks are distributed amongst you?
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II. Experiment (100 Pts) (Please discuss with your mentor about the expectations)

1. Implementation (50 Pts)

- a. Implement the novelties/feedback.
- b. Get the boiler code ready with the entire pipeline! This includes your dataset preparation, training, and evaluation for novelty.
 - i. Note: Do not reproduce the results for closed-source/larger LLMs.
- c. Feel free to use the existing code, update it, or write the code from scratch.

2. Reporting the results/findings (10 Pts)

- a. Report the pattern in results and findings from the pattern for the smaller LLMs/baselines.

3. Viva (40 Pts)

- a. TAs will conduct the viva for the report and code execution.
- b. You need to meet with your mentors regularly for weekly updates.